

What WER Hides: A Closer Look at Algerian Code-Switched ASR

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Abstract

Automatic speech recognition for Algerian social-media speech is hard because of dialectal variation, Arabic-French code-switching, and non-standard orthography, yet these difficulties are usually summarised by a single word error rate that hides where recognition fails and whether gains reach information-bearing content. We evaluate seven systems on roughly 30 minutes of Algerian political social-media speech, annotated by language variety as Algerian dialect, Modern Standard Arabic, code-switched, or French: a Whisper-large-v3 baseline, five Whisper variants fine-tuned on different combinations of broadcast, dialectal, and code-switched speech, and the prompted Qwen3-Omni model. The best system reduces overall WER from 60.1% to 31.3%, but still reaches 44.0% on code-switched speech, and named entities remain harder than ordinary tokens, with fewer than half preserved. These results show that ASR difficulty on Algerian social-media speech cannot be read from overall WER alone; it must be measured separately by language variety and by whether named entities survive.

1 Introduction

Automatic speech recognition (ASR) for Algerian Arabic is difficult for several reasons. Algerian Arabic differs substantially from Modern Standard Arabic (MSA) in lexicon, morphology, pronunciation, and syntax (Habash, 2010; Harrat et al., 2016; Talafha et al., 2024), and as a Maghrebi variety, it is among the Arabic dialects least intelligible to systems trained on MSA. Speakers alternate frequently between Algerian Arabic and French, often within a single utterance (Amazouz et al., 2017; Lachemat et al., 2024), and the dialect has no standardised orthography, appearing in either Arabic or Latin script (Habash et al., 2012; Zaidan and Callison-Burch, 2014; Besdouri et al., 2024). When this speech comes from social media, these difficulties

combine with casual, unscripted delivery, background noise, compression artefacts, and variable recording conditions.

Most ASR evaluations report only word error rate (WER), which hides differences between linguistic varieties and treats unlike errors alike. A system may transcribe MSA well while failing on Algerian dialect, recognise Arabic speech while forcing French expressions into Arabic script, or lower its overall WER while still corrupting the names of people, places, and organisations. WER counts each of these the same way, whether a word is harmlessly respelled or replaced by an unrelated one. Such failures matter most in political and misinformation-related content, where named entities determine who or what a claim concerns.

Arabic dialectal ASR has advanced largely through the MGB challenges (Ali et al., 2017, 2019) and large MSA broadcast corpora (Mubarak et al., 2021), with fine-tuned multilingual models such as Whisper now holding the best published results across dialects (Talafha et al., 2023, 2024). Progress on Algerian Arabic lags, however: zero-shot Whisper large-v2 reaches 103.4% WER on Algerian speech (Talafha et al., 2023), and the only open corpora for fine-tuning, CAFE (Lachemat et al., 2024) and the Algerian subset of Casablanca (Talafha et al., 2024), together provide barely four hours of training data, and recent shared tasks confirm it remains among the weakest-resourced Arabic varieties for ASR (Salhab et al., 2025; Elleuch et al., 2025). Code-switched ASR remains particularly underserved, since most resources transcribe foreign words in a single script and dedicated bilingual models for Arabic are scarce (Sitaram et al., 2019; Hamed et al., 2025). A parallel line of work has questioned whether WER reflects transcription quality on dialectal speech at all, given that valid spelling variants are penalised as errors (Ali et al., 2015; Besdouri et al., 2024), motivating evaluation that looks beyond a single metric.

This paper asks how difficult ASR remains on Algerian social-media speech after fine-tuning, a setting relevant to downstream tasks such as fake-news analysis. We compare a multilingual Whisper model, several Whisper fine-tuned on different speech resources, and a large prompted audio-language model, evaluating all of them on the same variety-annotated segments. The fine-tuning corpora differ in which dialects they cover and in whether French is written in Arabic or Latin script. We use these differences to show that the choice of training data changes not only how accurate a system is but the kind of errors it makes, ranging from harmless rewrites, such as normalising dialectal words to their MSA form, through transliterating or corrupting embedded French, to outright omission of difficult speech and hallucination of words with no acoustic support. Because these errors fall unevenly on the words that carry meaning, we further probe how well each system preserves named entities, the persons, organisations, and locations that determine who or what a claim concerns.

To support this study, we construct an evaluation set of Algerian political and misinformation-related social-media speech in which every segment is labelled with its language variety: Algerian dialect, Modern Standard Arabic, mixed MSA and dialect, Arabic-French code-switching, or French. These labels let us report performance per variety rather than as one overall figure, and to read the script and spelling behaviour behind it.

Our contributions are:

1. An evaluation set of Algerian political and misinformation-related social-media speech in which every segment is labelled with its language variety: dialect, MSA, mixed MSA and dialect, Arabic-French code-switching, or French.
2. A per-variety evaluation of seven systems showing that the strongest remains at 31.3% WER overall, 37.5% on dialectal and 44.0% on code-switched speech, and that systems fail in distinct ways, the weakest two by omission and by hallucination, respectively.
3. An analysis of fine-tuning data and output script showing that the gain is uneven: Arabic-script-only supervision improves Algerian speech while almost eliminating correct Latin-script output, whereas the mixed-script CAFE corpus prevents this collapse but does not

recover the baseline model’s monolingual-French strength.

4. A named-entity probe showing that entity transcription errors stay higher than general transcription errors, that person names are the hardest to preserve, and that fine-tuning changes which entities survive rather than uniformly improving them.

2 Data and Experimental Setup

2.1 Evaluation Set

We evaluated all systems on 335 manually transcribed audio segments extracted from Algerian political and misinformation-related social-media videos¹. Each segment is labelled with the language variety it contains: Algerian dialect (DI), mixed MSA and dialect (MSA-DI), Modern Standard Arabic (MSA), Arabic-French code-switching (CS), or French (FR). Table 1 reports the distribution of the evaluation set.

Variety	DI	MSA-DI	MSA	CS	FR	Total
Segments	74	127	83	19	32	335
Duration (min)	5.5	12.3	4.6	4.7	2.6	29.7

Table 1: Variety distribution and duration of the evaluation set.

The evaluation set comprises approximately 30 minutes of transcribed speech drawn from 10 social-media videos collected in September 2025, divided into 335 segments. Segments were obtained automatically using the SpeechBrain CRDNN voice-activity detector², with energy-based refinement, merging of segments closer than 0.25 seconds, and removal of segments shorter than 0.25 seconds. The resulting segments have a mean length of 5.1 seconds, a median of 3.7 seconds,

¹The corpus consists of audio segments extracted from publicly posted political and misinformation-related videos on TikTok and YouTube, collected for non-commercial academic research consistent with each platform’s terms of service. Speakers are public figures (politicians, journalists, and commentators) speaking in their public political capacity; no private individuals or non-public content were collected. The corpus is released, not the source videos, at <https://huggingface.co/datasets/oist/algerian-code-switched-asr-eval> under a controlled research-use agreement modelled on that of the BOUTEF corpus (Smaili et al., 2026), restricting use to non-commercial research, prohibiting redistribution, and prohibiting attempts to identify, contact, track, or profile speakers beyond what is already apparent from the recordings.

²<https://huggingface.co/speechbrain/vad-crdnn-libriparty>

and a range from 0.4 to 30.4 seconds. Transcription and variety labelling were carried out by a native speaker of Algerian Arabic fluent in French, familiar with the code-switching phenomena and the downstream task. Dialectal Arabic was written as heard, with no normalising standard; MSA followed standard Arabic orthography; and French was kept in Latin script following standard French orthography.

2.2 Fine-tuning Corpora

Table 2 summarises the four corpora used to fine-tune Whisper-large-v3 (Radford et al., 2023). Casablanca (Talafta et al., 2024) and CAFE (Lachemat et al., 2024) both contain Algerian Arabic-French code-switching but follow different transcription conventions. Casablanca represents embedded French words predominantly in Arabic script, providing Algerian acoustic and lexical supervision without teaching the model to emit Latin-script tokens. CAFE retains French and English words in their original Latin script, demonstrating that Arabic- and Latin-script tokens may occur within the same utterance.

MLS (Pratap et al., 2020) provides monolingual French read speech, while QASR (Mubarak et al., 2021) contains large-scale broadcast Arabic dominated by MSA. Together, the four corpora let us compare fine-tuning by dialect match, code-switching, script convention, and training-data scale.

The source videos for the evaluation set were collected in September 2025, after the release of both the CAFE and Casablanca corpora, and therefore do not overlap with the fine-tuning data. CAFE and the evaluation set share broad source characteristics and collection channels but differ in topic and communicative setting. We therefore treat this evaluation as a same-source but topically shifted real-world case study rather than a strictly out-of-domain benchmark, measuring whether general Algerian and code-switched speech resources transfer to information-dense political content.

2.3 ASR Systems

Table 3 summarises the seven systems. The baseline Wv3 is Whisper-large-v3 (Radford et al., 2023), a multilingual speech recognition model pretrained on a large weakly supervised corpus³,

³<https://huggingface.co/openai/whisper-large-v3>

used here with its original parameters and without task-specific fine-tuning. We do not force an Arabic language tag, so that the model can produce French where appropriate, rather than being pushed toward a single language on multilingual segments. The five fine-tuned systems all start from this same Whisper-large-v3 checkpoint and were trained with identical hyperparameters, differing only in their training data, as detailed in the table. Qwen3-Omni (Qwen/Qwen3-Omni-30B-A3B-Instruct⁴) is evaluated zero-shot, prompted with the following instruction:

“Transcribe this audio into text. The audio can be in Algerian Arabic, Modern Standard Arabic, French, or a mix of them.”

All fine-tuned systems share an identical training configuration: AdamW with learning rate 1e-5, a Noam scheduler with 500 warmup steps, batch size 4 with gradient accumulation factor 4 (effective batch size 16), 10 epochs with no early stopping, fp16 precision, and a maximum utterance length of 40s. Speed perturbation, frequency dropout, and chunk dropout are applied as data augmentation using the SpeechBrain toolkit (Ravanelli et al., 2021). All Whisper-based systems decode with beam search, beam size 8. Qwen3-Omni is prompted zero-shot via vLLM with temperature 0.2, top- p 0.95, top- k 20, and a maximum of 8,192 generated tokens. Table 4 reports the hours sampled per system.

2.4 Text Normalization and ASR Metrics

We report word error rate (WER) and character error rate (CER), computed with the jiwer toolkit⁵. Each reported figure is pooled over all segments in the relevant group rather than averaged across per-segment rates: the overall column pools all 335 segments, and each per-variety column pools the segments carrying that variety label. WER is defined as

$$\text{WER} = \frac{S + D + I}{N}, \quad (1)$$

where S , D , and I denote substitutions, deletions, and insertions, and N is the number of reference words. CER applies the same calculation at the character level.

All primary results are computed after applying the same deterministic normalisation to references

⁴<https://huggingface.co/Qwen/Qwen3-Omni-30B-A3B-Instruct>

⁵<https://github.com/jitsi/jiwer>

Corpus	Hours	Domain	Variety	Transcription convention
Casablanca (Algerian)	~2	YouTube/television speech	Algerian, CS	French in Arabic script
CAFE small v2	~2	Radio and podcast speech	Algerian, CS	French and English in Latin script
MLS French	1,076	Read speech	French	Latin script
QASR	2,000	Broadcast news	MSA, dialects	Standard Arabic script

Table 2: Fine-tuning corpora used in the evaluated systems. Reported sizes are full corpus sizes; the hours actually sampled in each run are given in Table 3.

System	Condition	Role in the analysis
Wv3	Whisper-large-v3, no forced language tag	multilingual baseline
Wv3-QASR	Fine-tuned on QASR	MSA-heavy broadcast
Wv3-CASA	Fine-tuned on Casablanca (Algerian)	Arabic-script Algerian
Wv3-CAFE	Fine-tuned on CAFE small v2	Mixed-script code-switched
Wv3-CC	Fine-tuned on Casablanca + CAFE	Combined Algerian, mixed-script
Wv3-ALL	Casablanca + CAFE + 2h each MLS, QASR	Algerian plus limited FR and MSA
Qwen3-Omni	Zero-shot prompted transcription	Instruction-tuned comparator

Table 3: Evaluated ASR systems and their role in the analysis.

System	Hours sampled
Wv3-QASR	2000
Wv3-CASA	~2
Wv3-CAFE	~2
Wv3-CC	~4
Wv3-ALL	~8

Table 4: Hours sampled per fine-tuning corpus for each system.

and hypotheses. The normalisation removes Arabic diacritics and tatweel, maps the hamzated alif forms to bare alif, maps hamza-on-waw and hamza-on-ya to their carrier letters, and maps alif maqsura to ya. It converts Arabic-Indic digits to Western digits, normalises curly apostrophes and backticks to a straight apostrophe, lowercases Latin text, removes unsupported punctuation and symbols, and collapses repeated whitespace.

2.5 Named-Entity Fidelity

We use named-entity recognition as a probe of whether named entities survive ASR. We apply the Arabic GLiNER model (Zaratiana et al., 2024) NAMAASpace/gliner_arabic-v2.1⁶ to the reference transcripts with a confidence threshold of 0.5 and retain person, location, and organization entities. These detected spans are treated as NER-derived reference entities rather than manually annotated gold entities, and the same set of reference entities is used to evaluate every ASR system.

We report four complementary measures. NER-WER is WER restricted to reference tokens covered by entity spans, which we compare with overall WER on the same entity-bearing segments. Entity-

CER is the character error rate between each reference entity and its aligned form in the ASR hypothesis, taken from the word-level reference-hypothesis alignment, with the entity boundary fixed by the reference transcript. The other two measures apply the same GLiNER model to each ASR hypothesis and ask whether each reference entity is recovered: surface recall requires agreement on the normalized entity surface but not its type, while typed recall additionally requires agreement on the entity type. Because they re-run NER on noisy hypotheses, surface and typed recall reflect the NER model’s robustness as well as ASR quality, and are read as downstream measures.

3 Evaluating ASR on Code-Switched Algerian Speech

This section evaluates the systems described in Section 2 on the variety-annotated evaluation set. Our main comparison is between four of them: the unforced Whisper v3 baseline (Wv3); Wv3-QASR, fine-tuned on the large MSA-heavy QASR corpus; Wv3-CC, fine-tuned on the Casablanca and CAFE Algerian code-switched data; and the zero-shot Qwen3-Omni. These four span the main axes of interest: a baseline, a broadcast MSA fine-tuning, a dialect-matched code-switched fine-tuning, and a prompted multilingual model. The remaining systems (Wv3-CASA, Wv3-CAFE, and Wv3-ALL) are introduced in the data ablation and the error analysis, where they isolate the contribution of each fine-tuning corpus.

⁶https://huggingface.co/NAMAASpace/gliner_arabic-v2.1

Model	ALL	CS	DI	FR	MSA	MSA DI
<i>WER (%)</i>						
Wv3-CC	31.3	44.0	37.5	45.7	23.5	28.6
Qwen3-Omni	43.4	52.8	62.5	27.3	41.0	30.5
Wv3-QASR	59.1	77.5	69.9	100.0	50.9	30.5
Wv3	60.1	92.8	125.2	25.9	40.1	39.1
<i>CER (%)</i>						
Wv3-CC	14.4	25.6	13.7	34.0	8.5	11.3
Qwen3-Omni	20.8	34.9	26.8	19.2	17.6	13.3
Wv3-QASR	40.3	65.3	41.7	95.6	30.0	12.8
Wv3	37.0	76.2	89.7	13.9	15.8	21.7

Table 5: WER and CER on the evaluation set by variety, as percentages on normalised text. Bold marks the best score per column within each metric. On CS and FR, several bolded differences are not statistically distinguishable given the small segment counts ($n=19$, $n=32$); see the accompanying text and Appendix A.

3.1 Quantitative Evaluation

Table 5 reports WER and CER for the four systems (see Appendix A for details about confidence intervals). The overall ranking is led by Wv3-CC, but the per-variety columns are more informative, since each system’s strengths track the data it was trained on. Wv3-CC is far ahead of Wv3-QASR on code-switched speech (44.0% vs 77.5% WER, with non-overlapping 95% bootstrap intervals of [33.2, 58.5] and [68.1, 84.9]). Its apparent advantage over Qwen3-Omni (52.8%, interval [39.8, 71.8]) and Wv3 (92.8%, interval [50.0, 168.0]) is less certain: with only 19 code-switched segments, these intervals overlap Wv3-CC’s, so the true ranking among these three systems on this variety cannot be confirmed from this evaluation set (Appendix A). Its fine-tuning data is the only one to include code-switched speech with French kept in Latin script, so it alone learns to alternate scripts within a segment. On the pure-French subset, the result reverses, with Wv3 best (25.9% WER, interval [18.5, 36.4]) and Wv3-CC the weakest fine-tuned system (45.7%, interval [29.8, 67.3]); the intervals overlap, so with only 32 French segments this gap is not clearly distinguishable at the segment level (Appendix A). This follows from the same data: the Algerian fine-tuning set contains no monolingual French, only French embedded in code-switched Algerian segments, so the model never learns French as a standalone language the way Whisper’s pretraining or Qwen3-Omni’s multilingual training do. Wv3-QASR shows the mirror image: fine-tuned on 2,000 hours of mostly-MSA broadcast Arabic, it is competitive on MSA (30.5% WER, level with Qwen3-Omni) but does much

worse on code-switched (77.5%) and pure-French (100.0%) speech, since its data is overwhelmingly monolingual MSA. Wv3-CC is strongest on dialect (37.5% WER) and on the MSA-dialect combination (23.5%), where its training data fits best.

We interpret the relationship between WER and CER by considering their ratio rather than their difference. CER is everywhere lower than WER, but the ratio varies. For the dialect-fine-tuned systems on dialectal speech, it is well under one half (CER 13.7% against WER 37.5% for Wv3-CC, 26.8% against 62.5% for Qwen3-Omni), whereas for the weaker systems, the two metrics stay closer together. A low ratio means many word errors differ from the reference by only a few characters. Because Algerian dialect has no standardised orthography, these are often valid alternative spellings of the reference rather than different words: WER counts them as full errors while CER registers only a partial difference. This is consistent with prior work showing that WER over-penalises orthographic variation in dialectal Arabic and morphologically rich languages, and that CER tracks human judgements of transcription quality more closely (Ali et al., 2015; Thennal et al., 2025). We do not treat the ratio as a general diagnostic across all conditions: on French, for example, CER stays close to WER because the errors are genuine language and script failures rather than spelling variants. We examine the nature of these dialectal spelling variants in the qualitative analysis.

3.2 Where the Gain Comes From

System	ALL	DI	CS	FR	MSA
Wv3-CASA	51.2	50.6	66.2	100.9	40.8
Wv3-CAFE	36.6	49.2	49.4	35.6	29.5
Wv3-CC	31.3	37.5	44.0	45.7	28.6
Wv3-ALL	39.4	52.3	51.1	41.4	31.6

Table 6: Data ablation (WER %, normalised). All systems fine-tune the same Whisper v3 base, varying only the fine-tuning corpora. Bold marks the point-estimate best per column; on CS, differences among Wv3-CC/CAFE/CASA/ALL are not statistically distinguishable given $n=19$ (Appendix A).

To see which fine-tuning data drives the improvement, we compare four systems built on the same Whisper v3 base (Table 6). Casablanca alone (Wv3-CASA) gives the weakest result: it improves little on dialect (50.6% WER) and, because its French is written in Arabic script, it learns to emit only Arabic and reaches 100.9% WER on pure French,

effectively unable to produce Latin-script output. CAFE alone (Wv3-CAFE), with the same amount of data, reaches 36.6% overall and keeps French usable (35.6%), since its training French is in Latin script. Combining the two (Wv3-CC) is best overall (31.3%), with each corpus covering the other’s weakness: Casablanca contributes dialectal coverage and CAFE the Latin-script code-switching. Wv3-ALL adds a small amount of MLS French and QASR, kept limited so as not to push the model toward French or MSA. Its French figure (41.4%, interval [28.6, 59.1]) is not statistically distinguishable from Wv3-CC’s (45.7%, interval [29.8, 67.3]), but it lowers dialect accuracy (52.3% against 37.5%) and is not better overall (39.4% against 31.3%), both clear differences with non-overlapping intervals (Appendix A); its MSA figure (31.6% against 28.6%) is likewise not distinguishable from Wv3-CC. How the proportion of each corpus in the mixture shapes this balance, and whether a different ratio could add French and MSA coverage without reducing dialect accuracy, is a question this result raises for future work.

3.3 Qualitative Error Analysis

A qualitative review of the transcriptions complements the quantitative results and explains several of the patterns above.

The first error mode is dialectal normalisation. The MSA-fine-tuned systems, such as Wv3-QASR, often map dialectal forms to their Modern Standard equivalents even when the dialectal word is recognised correctly, for example, rewriting “خمسطاش نيوم” (*khmstas nyoum*, fifteen days) as “يوم 15” and the dialectal “عندو” and “تتكمو” as the standard “عنده” and “تتكلم”. Dialect-fine-tuned systems such as Wv3-CC retain the original surface forms more often, with only minor phonetic-orthographic variation (e.g. “فطيحه” for “فضيحة”, both *fadhiha*, scandal). These normalisations are the spelling variants explaining the large WER/CER gap noted earlier: WER counts them as full errors, although the content is preserved, while CER registers them as only partial differences.

The second error mode concerns code-switching between Arabic and French. An embedded French word is either kept in Latin script or transliterated into Arabic, depending on the fine-tuning data: only data containing Latin-script French within Algerian speech, namely CAFE, teaches a model

System	S ↓	D ↓	I ↓	H ↑
Wv3-CC	865	152	97	2132
Wv3-QASR	1000	1049	55	1365
Qwen3-Omni	1136	313	94	1858
Wv3	1181	336	622	1872

Table 7: Error-type counts: substitutions (S), deletions (D), insertions (I), and hits (H). Arrows show the preferred direction; bold marks the best per column.

to alternate scripts within a segment, and without it, a model defaults to a single script. On a segment naming the ranks “inspecteur général” and “divisionnaire général”, the CAFE-fine-tuned systems (Wv3-CAFE and Wv3-CC) keep them in Latin, while the others transliterate or degrade them. Transliteration is not always harmless: a faithful one preserves the sound and leaves the word recoverable (“transfert” → *ترونسفار*), but in other cases the embedded word is mapped onto an unrelated Arabic word and the meaning is lost, as when “divisionnaire général” becomes “تلفزيون جنرال” (*tilfizyoun jeneral*, television general) or “au maroc” becomes “و معروف” (*wa maarouf*, and well-known). WER and CER both count the faithful and the meaning-changing version as the same error, so the difference between a recoverable and an unrecoverable word is invisible to the aggregate scores.

Finally, catastrophic language misidentification occurs on pure French segments for the systems most conditioned on Arabic, such as Wv3-QASR, which emit hallucinated Arabic and reach WER values at or above 100%. The base Wv3 is the opposite case, transcribing pure French accurately without a forced language tag, though its overall WER stays high because of instability on code-switched and dialectal input. Taken together, the errors range from harmless MSA normalisation through meaning-changing substitution in code-switching to outright language misidentification, which is why aggregate WER and CER, sensitive as they are to orthography and formatting (Besdouri et al., 2024), do not fully reflect transcription quality on dialectal and code-switched speech.

3.4 Error Type Distribution

To characterise how the systems fail, we decompose each one’s errors into substitutions, deletions, insertions, and correctly transcribed words, or hits (Table 7). The two weakest systems fail in opposite ways. Wv3-QASR is the most conservative, with

the fewest insertions (55) but by far the most deletions (1049): conditioned on MSA, it omits hard dialectal and code-switched stretches rather than guessing. Wv3 fails the opposite way, producing the most insertions (622), the signature of hallucination on dialectal and code-switched input, where it generates tokens with no acoustic support, which is also why its WER exceeds 100% on some varieties. Wv3-CC sits between these extremes, with the fewest substitutions (865) and deletions (152) and the most hits (2132), a balanced and faithful alignment with the reference. Qwen3-Omni is intermediate, with moderate insertions and the second-highest hit count. This contrast between omission and hallucination, rather than a single error rate, is what dialect-matched, code-switched fine-tuning resolves.

4 Named Entities as a Difficulty Probe

Our goal is not to evaluate named-entity recognition itself. Instead, we use named entities as a targeted probe of ASR quality. Named entities are lexically sparse, topic-dependent, and information-bearing. Errors affecting a person, organization, or location may change the target of a political claim even when the surrounding utterance remains recognizable.

As described in Section 2.5, we apply the same GLiNER model to the reference transcripts to obtain NER-derived reference entities, and evaluate every system with the four measures defined there: NE-WER, Entity-CER, surface recall, and typed recall.

4.1 Overall Entity Fidelity

System	CER	WER	Typed rec.	Surface rec.
Wv3	38.1	48.7	35.5	38.6
Wv3-QASR	38.7	61.0	27.7	29.0
Wv3-CASA	33.8	44.4	33.0	36.9
Wv3-CC	26.3	35.5	46.6	48.9
Wv3-CAFE	29.8	38.9	41.4	44.5
Wv3-ALL	30.7	40.9	41.0	43.5
Qwen3-Omni	35.0	48.3	34.6	36.2

Table 8: Named-entity fidelity on normalised transcripts, as percentages. CER and WER denote Entity-CER and NE-WER; the four measures are defined in Section 2.5.

Table 8 shows that Wv3-CC obtains the strongest result under every entity measure. It reduces entity-CER from 38.1% to 26.3% and NE-WER from 48.7% to 35.5% relative to the Wv3 baseline. Its surface recall reaches 48.9%, an improvement of 10.3 points over the baseline.

Nevertheless, even the strongest system fails to preserve more than half of the NER-derived reference entities. This residual error is operationally important for political and misinformation-related speech, where corruption of a person, organization, or location may change who or what a claim concerns.

Surface recall is only between 1.3 and 3.9 points higher than typed recall across systems. For Wv3-CC, the difference is 2.3 points. Most entity loss, therefore, arises from failure to recover the entity surface itself rather than only from assigning an incorrect entity type after transcription.

The CAFE-based systems obtain the strongest entity-level results. In contrast, Wv3-QASR reaches 61.0% NE-WER and only 29.0% surface recall, indicating that fine-tuning on a large MSA-heavy broadcast corpus does not provide robust entity preservation in the target setting.

4.2 Which Entities Are Hardest?

System	LOC	ORG	PER
Wv3	57.8	37.4	27.7
Wv3-QASR	46.6	30.6	17.6
Wv3-CASA	49.1	34.7	30.7
Wv3-CC	60.2	50.3	41.2
Wv3-CAFE	60.2	40.1	37.5
Wv3-ALL	60.2	43.5	33.3
Qwen3-Omni	55.3	40.8	22.1

Table 9: Surface recall (%) by reference entity type. LOC, ORG, and PER denote location, organization, and person entities, respectively. Bold indicates the highest value in each column.

Table 9 presents the Surface recall performance for the three types of entities: locations (LOC), organizations (ORG), and persons (PER). Person entities have the lowest surface recall for every evaluated system. Under the baseline, only 27.7% of person entities survive, compared with 57.8% of locations and 37.4% of organizations. Wv3-CC improves person recall to 41.2%, but still fails to preserve most person names.

Locations are consistently the easiest entity type. Wv3-CC, Wv3-CAFE, and Wv3-ALL each recover approximately 60% of location surfaces, whereas no system recovers more than half of person entities. Entity difficulty is therefore not uniform: person names remain the principal residual entity-level failure.

Entity-CER shows a consistent pattern for most systems. For example, under Wv3, entity-CER is 43.4% for persons, compared with 33.6% for both

locations and organizations. Under Qwen3-Omni, person entity-CER reaches 39.6%, compared with 28.8% for locations and 33.8% for organizations. Wv3-CC substantially narrows this type-specific gap, obtaining entity-CER values of 26.7% for locations, 25.6% for organizations, and 26.5% for persons.

4.3 Entity Difficulty Across Varieties

Variety	Entity-CER ↓	Surface recall ↑
CS	50.1	26.3
DI	23.5	41.4
FR	43.1	25.6
MSA	18.8	64.0
MSA-DI	22.7	56.4

Table 10: Entity fidelity of Wv3-CC by linguistic variety.

Table 10 shows that the variety asymmetry observed under general WER also appears in entity preservation. Wv3-CC obtains its lowest entity-CER on MSA at 18.8%, followed by MSA-DI at 22.7% and DI at 23.5%. Entity-CER increases to 43.1% on French and 50.1% on code-switched speech.

The same contrast appears under surface recall. Wv3-CC preserves 64.0% of MSA entities and 56.4% of MSA-DI entities, but only 41.4% of dialectal entities and approximately one quarter of entities in French and code-switched segments. Entity preservation, therefore, sharpens the variety-level conclusion: the strongest system remains particularly unreliable when names occur in code-switched or monolingual French speech.

4.4 Fine-tuning Changes Which Entities Survive

Overall entity scores do not show whether a system improves the same entity instances as the baseline. We therefore compare surface recovery decisions against Wv3.

Table 11 presents the number of *rescued* and *newly lost* entities by evaluated systems in comparison with the base Wv3. A reference entity is considered *rescued* when it is not preserved by Wv3 but is preserved by the evaluated system. It is considered *newly lost* when it is preserved by Wv3 but not by the evaluated system.

Wv3-CC rescues 121 entity instances missed by the baseline while newly losing 63, resulting in a net gain of 58 preserved entities. Wv3-CAFE and

System	Rescued	Newly lost	Net
Wv3-QASR	38	94	−56
Wv3-CASA	65	74	−9
Wv3-CC	121	63	+58
Wv3-CAFE	78	46	+32
Wv3-ALL	79	52	+27
Qwen3-Omni	45	64	−19

Table 11: Entity surfaces rescued and newly lost relative to the Wv3 baseline.

Wv3-ALL also produce positive net gains of 32 and 27 entities, respectively.

The effect of fine-tuning is not uniformly positive. Wv3-QASR rescues only 38 entities while newly losing 94, producing a net loss of 56. Qwen3-Omni also loses more baseline-preserved entities than it rescues. Fine-tuning, therefore, changes which entities survive, rather than simply applying a uniform reduction to entity error.

This instance-level view reinforces the corpus-level findings. Fine-tuning that includes mixed-script code-switched speech (CAFE) yields a net gain in preserved entities, while Arabic-script-only fine-tuning produces a net loss: Wv3-QASR and Wv3-CASA discard more entities than the multilingual baseline already recovered than they rescue, consistent with their bias toward a single output script.

4.5 Complementarity with Overall Error

System	Pearson	Spearman	R^2	NE:overall
Wv3	0.32	0.66	0.10	1.1×
Wv3-QASR	0.63	0.62	0.39	1.1×
Wv3-CASA	0.75	0.64	0.57	1.0×
Wv3-CC	0.76	0.58	0.57	1.2×
Wv3-CAFE	0.44	0.56	0.20	1.2×
Wv3-ALL	0.59	0.55	0.35	1.2×
Qwen3-Omni	0.75	0.63	0.56	1.2×

Table 12: Relationship between general transcription error and entity error. Pearson, Spearman, and R^2 are segment-level relationships between overall CER and entity-CER. NE: overall is the ratio of NE-WER to overall WER on the same entity-bearing segments.

We now examine how closely entity error tracks general transcription error (Table 12). Entity fidelity broadly follows general recognition difficulty: overall CER and entity-CER are positively correlated for every system, with Pearson correlations between 0.32 and 0.76 and Spearman correlations between 0.55 and 0.66, and the rankings induced by overall WER and by entity-CER agree closely (Spearman 0.93).

The relationship is nevertheless incomplete.

Segment-level R^2 values range from 0.10 to 0.57, so general character accuracy explains only part of the variation in entity fidelity, and the association is particularly weak for Wv3 and Wv3-CAFE. A lower overall WER does not guarantee better entity preservation: Qwen3-Omni reduces overall WER relative to Wv3, yet its surface recall falls from 38.6% to 36.2%, and Wv3-CASA improves dialectal recognition while its entity recall stays below the baseline.

5 Conclusion

We evaluated seven systems on Algerian political social-media speech, labelled by variety, to ask how difficult ASR remains after fine-tuning. Fine-tuning on Algerian code-switched data lowers overall WER from 60.1% to 31.3%, but the gain is uneven: the best system still misses more than a third of dialectal and nearly half of code-switched words, and the choice of fine-tuning corpus decides not only accuracy but the kind of errors a model makes. Our qualitative and error-type analyses show that these range from harmless dialectal respellings, through French transliterated or corrupted into Arabic, to omission of hard speech by the MSA-fine-tuned system and hallucination by the baseline. Using named entities as a probe, we found that names stay harder than ordinary tokens, that person names survive least often, and that fine-tuning changes which entities are preserved rather than improving them uniformly. A single WER number, therefore, cannot capture how these systems perform on Algerian code-switched speech: the per-variety results and the named-entity probe each reveal failures it hides, and the qualitative analysis tells a story the numbers alone cannot.

Limitations

Our evaluation set is small: about 30 minutes from ten videos in a single political domain, so results may not transfer to other domains, speakers, or conditions, and the smaller varieties (code-switched and French) rest on few segments. Transcripts and variety labels come from a single annotator, so we report no inter-annotator agreement. Because Algerian Arabic has no standard orthography, WER and CER penalise valid spelling variants as errors, so the scores overstate how often a word is genuinely wrong; our qualitative analysis shows examples of these harmless spelling variants and faithful transliterations alongside the meaning-changing substitu-

tions that actually lose content. The named-entity analysis uses entities detected by a single Arabic NER model rather than gold annotations, so its boundaries, types, and recall inherit that model’s errors. Finally, we test only Whisper-large-v3 variants and one prompted model under fixed hyperparameters, and other architectures or data mixtures may behave differently.

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A Bootstrap Confidence Intervals

Table 13 and Table 14 report the full per-variety WER and CER results with 95% bootstrap confidence intervals, computed as described in Section 2.4: segments are resampled with replacement 2,000 times within each variety group, and the pooled error rate is recomputed on each resample. These intervals support the claims made in Section 3 about which differences are, and which are not, statistically distinguishable.

System	ALL	CS	DI	FR	MSA	MSA-DI
Wv3	60.1 [45.6, 80.4]	92.8 [50.0, 168.0]	125.2 [62.4, 227.9]	25.9 [18.5, 36.4]	39.1 [31.5, 47.6]	40.1 [36.4, 43.6]
Wv3-QASR	59.1 [54.9, 63.0]	77.5 [68.1, 84.9]	69.8 [62.9, 76.4]	100.0 [100.0, 100.0]	30.5 [24.6, 36.9]	50.9 [45.7, 56.1]
Wv3-CASA	51.2 [47.3, 55.3]	66.2 [49.7, 87.2]	50.6 [44.4, 57.4]	100.9 [94.5, 105.4]	40.8 [34.5, 47.2]	40.4 [36.4, 44.2]
Wv3-CAFE	36.6 [33.7, 39.5]	49.4 [39.5, 65.7]	49.2 [43.9, 54.9]	35.6 [23.8, 52.1]	29.5 [23.6, 35.5]	30.9 [27.4, 34.4]
Wv3-CC	31.3 [28.3, 34.3]	44.0 [33.2, 58.5]	37.5 [31.3, 44.3]	45.7 [29.8, 67.3]	28.6 [23.0, 34.5]	23.5 [20.3, 26.9]
Wv3-ALL	39.4 [36.3, 42.4]	51.1 [39.4, 68.9]	52.3 [47.2, 58.0]	41.4 [28.6, 59.1]	31.6 [25.5, 38.3]	33.5 [30.1, 36.9]
Qwen3-Omni	43.4 [39.9, 46.9]	52.8 [39.8, 71.8]	62.5 [56.6, 68.4]	27.3 [17.1, 42.6]	30.5 [24.1, 37.1]	41.0 [37.3, 45.0]

Table 13: WER (%) with 95% bootstrap confidence intervals over 2,000 resamples, drawing segments with replacement and recomputing the error rate over the resampled set. CS ($n=19$) and FR ($n=32$) intervals are wide and frequently overlap across systems; differences on these two varieties should be read as suggestive rather than confirmed.

System	ALL	CS	DI	FR	MSA	MSA-DI
Wv3	36.9 [22.7, 56.0]	76.2 [30.7, 147.0]	89.7 [27.4, 187.5]	13.8 [8.8, 22.0]	21.7 [15.2, 29.3]	15.8 [13.7, 18.2]
Wv3-QASR	40.3 [35.1, 45.0]	65.3 [51.0, 74.7]	41.7 [33.6, 49.5]	95.6 [94.1, 96.7]	12.8 [9.3, 16.8]	30.0 [23.7, 36.3]
Wv3-CASA	25.3 [21.6, 29.6]	40.4 [27.3, 59.1]	18.1 [15.0, 21.7]	87.7 [81.0, 92.6]	15.8 [12.1, 20.1]	13.8 [12.0, 15.5]
Wv3-CAFE	15.6 [13.7, 17.6]	26.0 [18.9, 37.1]	19.1 [15.7, 22.7]	22.8 [12.6, 38.6]	12.9 [8.9, 17.3]	10.9 [9.3, 12.5]
Wv3-CC	14.4 [12.3, 16.4]	25.6 [18.7, 35.7]	13.7 [10.5, 17.2]	34.0 [20.5, 54.3]	11.3 [8.4, 14.6]	8.5 [7.2, 10.0]
Wv3-ALL	17.0 [15.0, 19.1]	29.7 [20.9, 43.0]	19.3 [16.2, 23.1]	27.5 [17.6, 42.5]	12.5 [9.0, 16.6]	11.9 [10.4, 13.5]
Qwen3-Omni	20.8 [18.5, 23.3]	34.9 [24.9, 50.1]	26.8 [23.1, 30.9]	19.2 [10.0, 33.5]	13.3 [9.6, 17.7]	17.6 [15.2, 20.1]

Table 14: CER (%) with 95% bootstrap confidence intervals over 2,000 resamples, drawing segments with replacement and recomputing the error rate over the resampled set. CS ($n=19$) and FR ($n=32$) intervals are wide and frequently overlap across systems; differences on these two varieties should be read as suggestive rather than confirmed.